

TuniCars+

Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

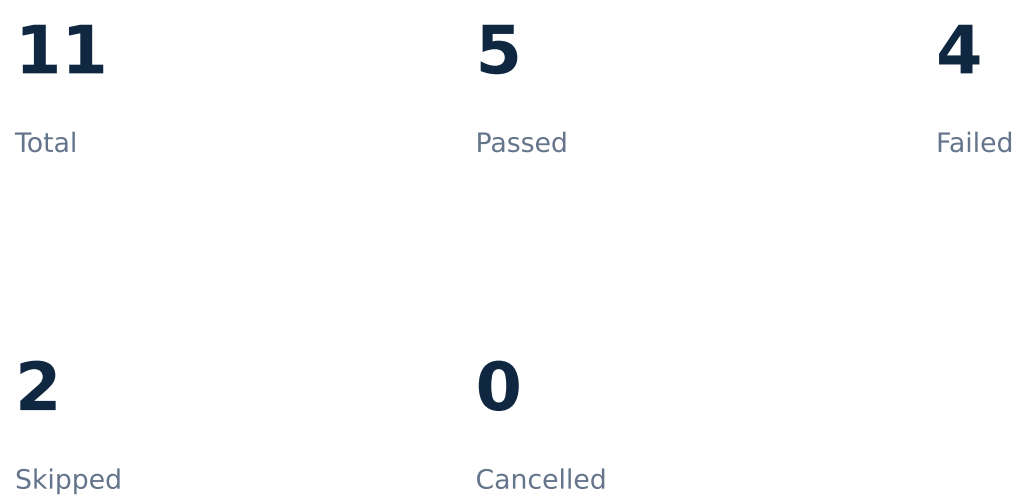
20260813_024830_127c0b

Date

2026-08-13T01:48:30.385082+00:00

Executive Summary

20260813_024830_127c0b



Overall measured availability: 47.70%

During this campaign, 5 of 11 scenarios passed, 4 failed and 2 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

Test Environment

Architecture and technologies

Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

Test Scenarios

Scenario	kubernetes	swarm
container-kill	1	1
cpu	1	1
latency	N/A	1
memory	1	1
network-partition	1	1
node-failure	1	1

Full Results Table

Scenario	Platform	Detection	Recovery	Required	Success	Avg ms	P95 ms	HTTP Fail	Avail. %	Status	Skip Reason	Error Message
container-kill	swarm	2.922	33.593	True	True	6.939	20.365	31	8.82	PASS	N/A	N/A
node-failure	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED	N/A Single-node Docker Swarm cannot demonstrate real node failure	N/A
cpu	swarm	N/A	N/A	N/A	Not Required	9.741	21.082	N/A	100.0	PASS	N/A	N/A
memory	swarm	N/A	N/A	N/A	Not Required	12.119	23.079	N/A	100.0	PASS	N/A	Critical cleanup / restoration failure: Swarm backend service did not return
network-partition	swarm	0.141	0.062	True	N/A	25.201	107.276	15	N/A	FAIL	N/A	Did not return
latency	swarm	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	FAIL	N/A	failed: Swarm backend health check HTTP 300
container-kill	kubernetes	8.547	19.203	True	True	1527.904	2043.33	6	25.0	PASS	N/A Kubernetes inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL	N/A
node-failure	kubernetes	N/A	N/A	N/A	N/A	Not Measured	Not Measured	N/A	Not Measured	SKIPPED		N/A
cpu	kubernetes	N/A	N/A	N/A	Not Required	91.385	398.115	N/A	100.0	PASS	N/A	N/A No healthy backend response was observed
memory	kubernetes	95.562	101.125	True	N/A	1418.356	4045.261	83	47.8	FAIL	N/A	was observed
network-partition	kubernetes	1.203	0.484	True	N/A	430.461	2208.605	12	N/A	FAIL	N/A	failure: Current scenario cancelled by user.

swarm — container-kill

State: **Pass**
Detection Time (s): 2.922

Recovery Time (s): 33.593

HTTP Success: N/A

HTTP Failures: 31

Availability (%): 8.82

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — node-failure

State: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Single-node Docker Swarm cannot demonstrate real node rescheduling.

swarm — cpu

State: Pass
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — memory

State: Pass
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): 100.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

swarm — network-partition

State: Fail
Detection Time (s): 0.141

Recovery Time (s): 0.062

HTTP Success: N/A

HTTP Failures: 15

Availability (%): N/A

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Critical cleanup/restoration failure: Swarm backend service did not return to Running state before timeout.

swarm — latency

State: **Fail**
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Precondition failed: Swarm backend health is not HTTP 200.

kubernetes — container-kill

State: Pass
Detection Time (s): 8.547
Recovery Time (s): 19.203

HTTP Success: N/A

HTTP Failures: 6

Availability (%): 25.0

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — node-failure

Status: SKIPPED
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: N/A

Availability (%): Not Measured

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Minikube inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL path after rescheduling.

kubernetes — cpu

Status: Pass
Detection Time (s): N/A
Recovery Time (s): N/A
HTTP Success: N/A
HTTP Failures: N/A
Availability (%): 100.0
CPU (%): N/A
Memory (MB): N/A

Recorded interpretation

No error or limitation recorded.

kubernetes — memory

State: **Fail**
Detection Time (s): 95.562

Recovery Time (s): 101.125

HTTP Success: N/A

HTTP Failures: 83

Availability (%): 47.8

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

No healthy backend response was observed during the stress test.

kubernetes — network-partition

State: Fail
Detection Time (s): 1.203

Recovery Time (s): 0.484

HTTP Success: N/A

HTTP Failures: 12

Availability (%): N/A

CPU (%): N/A

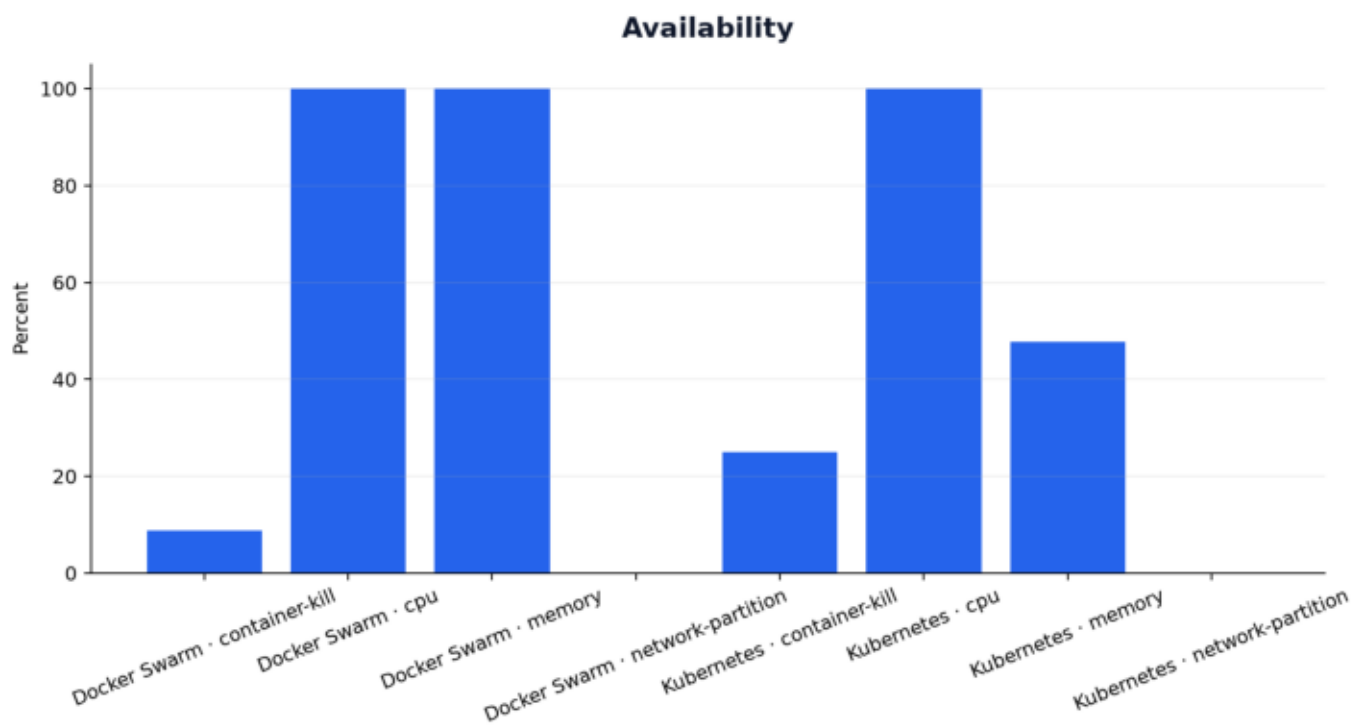
Memory (MB): N/A

Recorded interpretation

Critical cleanup/restoration failure: Current scenario cancelled by user.

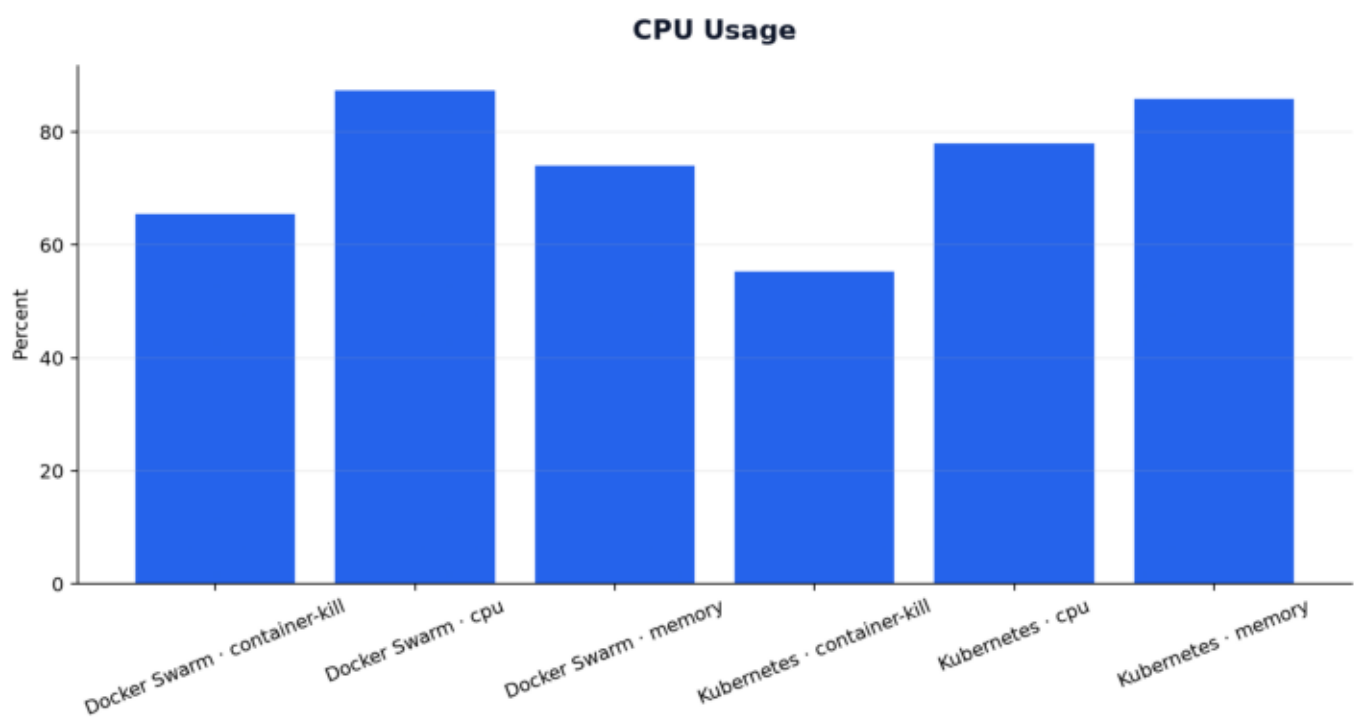
Availability

Current-session measured data



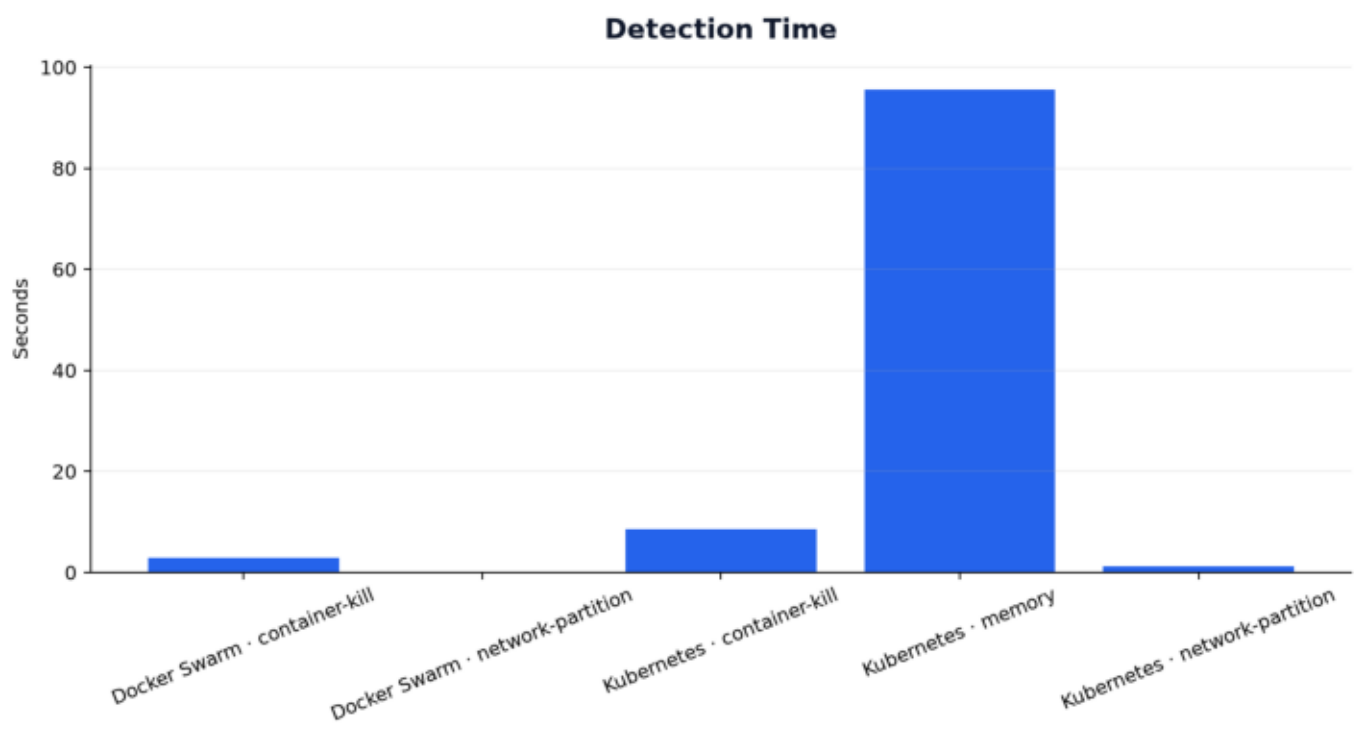
Cpu Usage

Current-session measured data



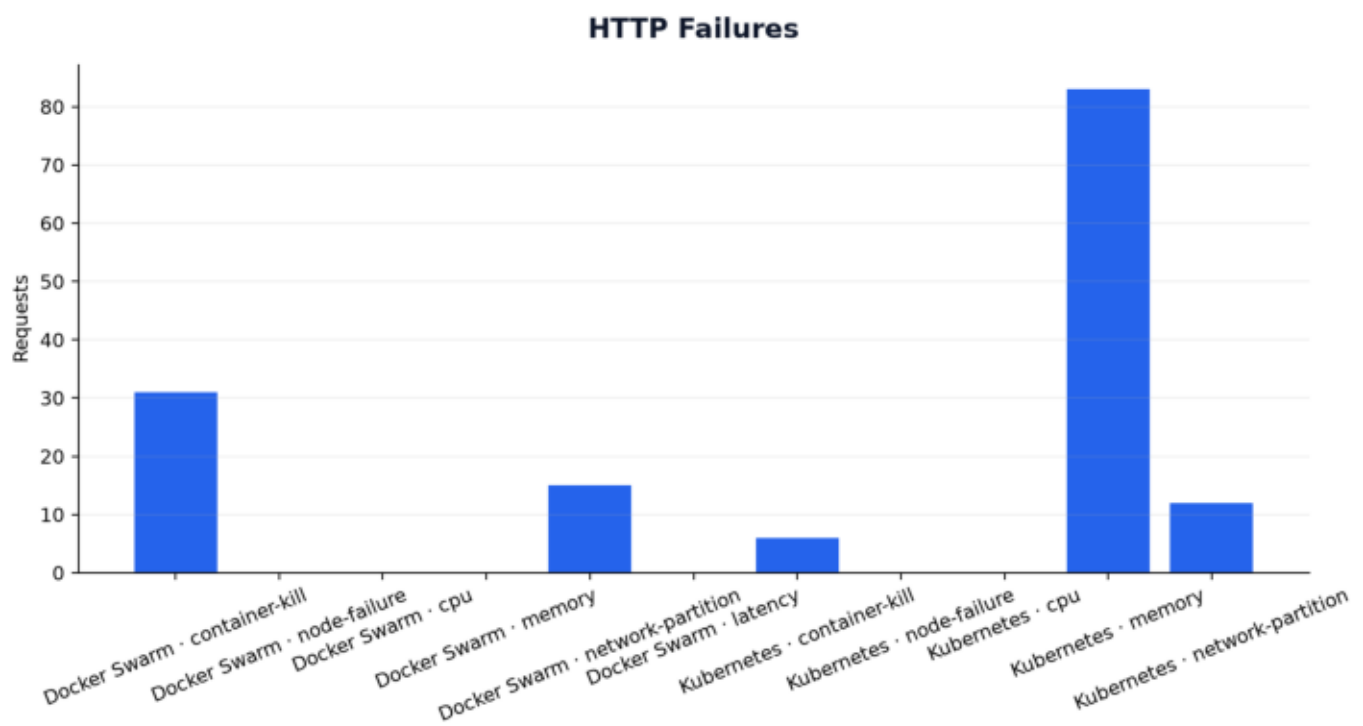
Detection Time

Current-session measured data



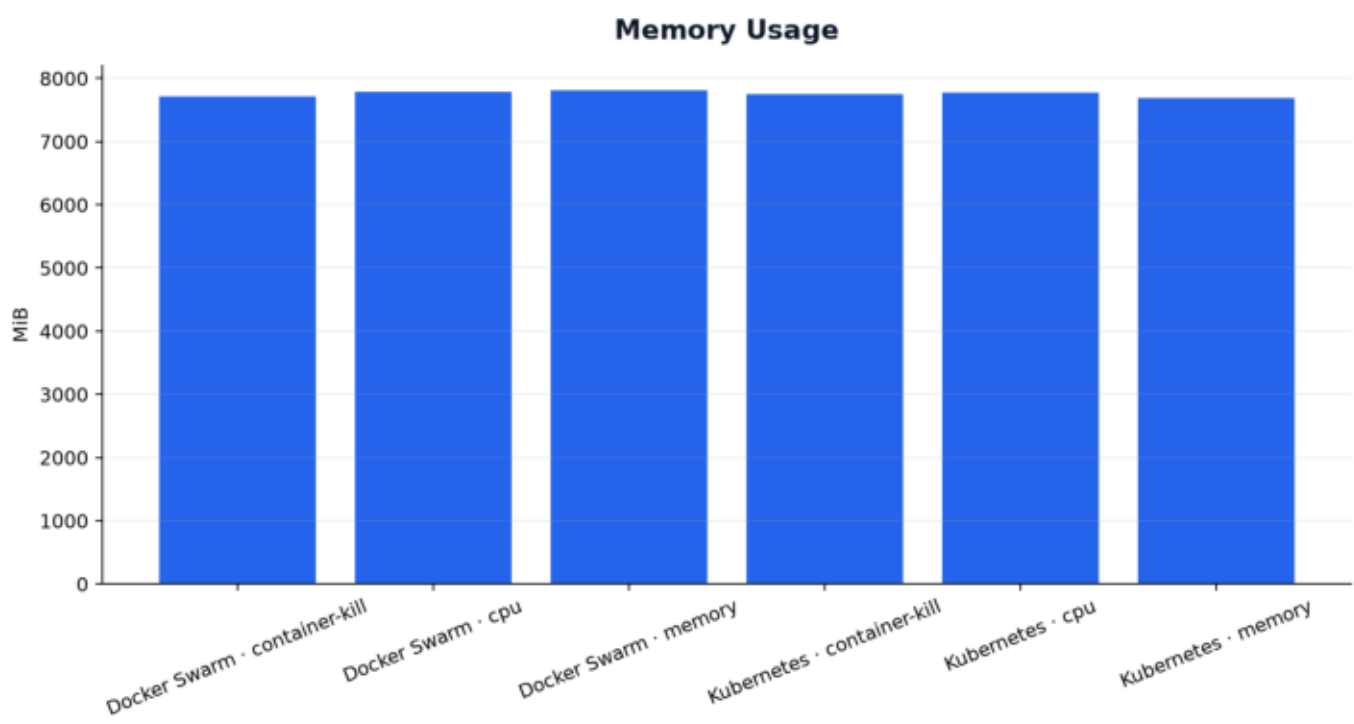
Http Failures

Current-session measured data



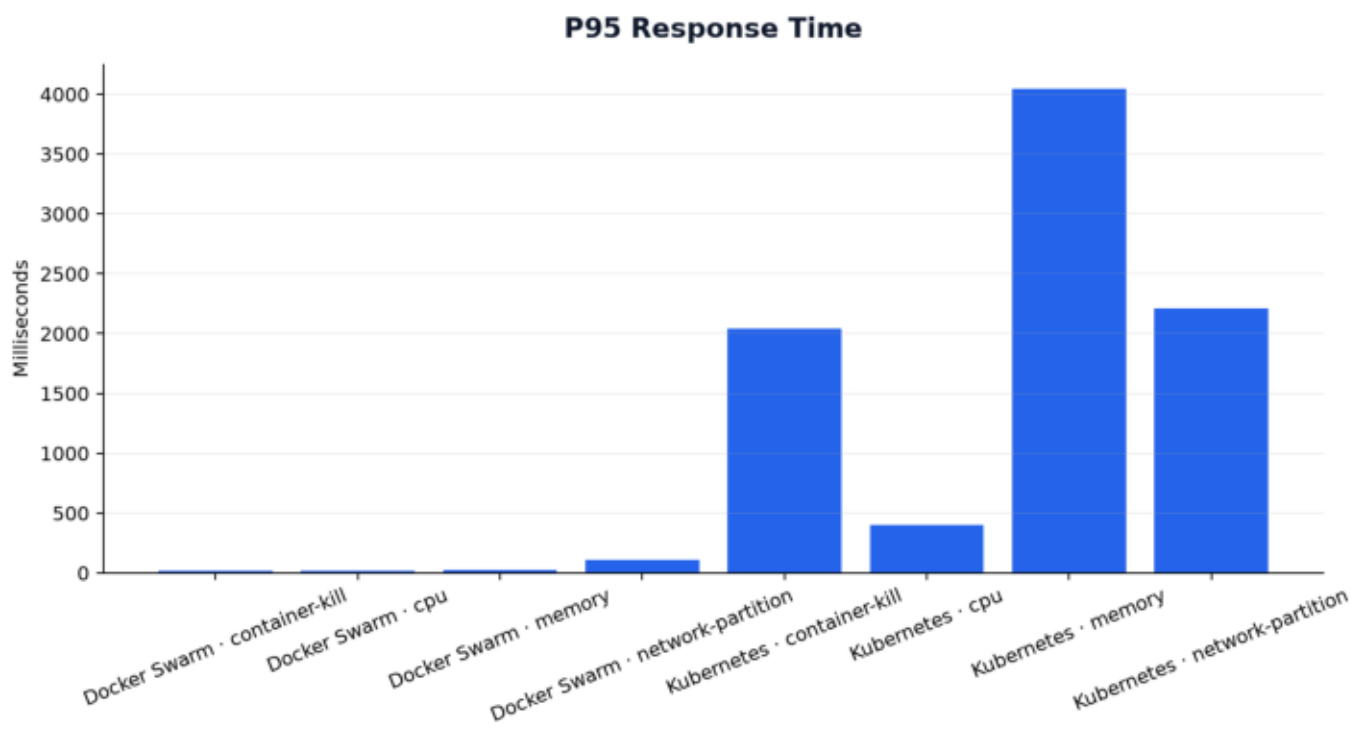
Memory Usage

Current-session measured data



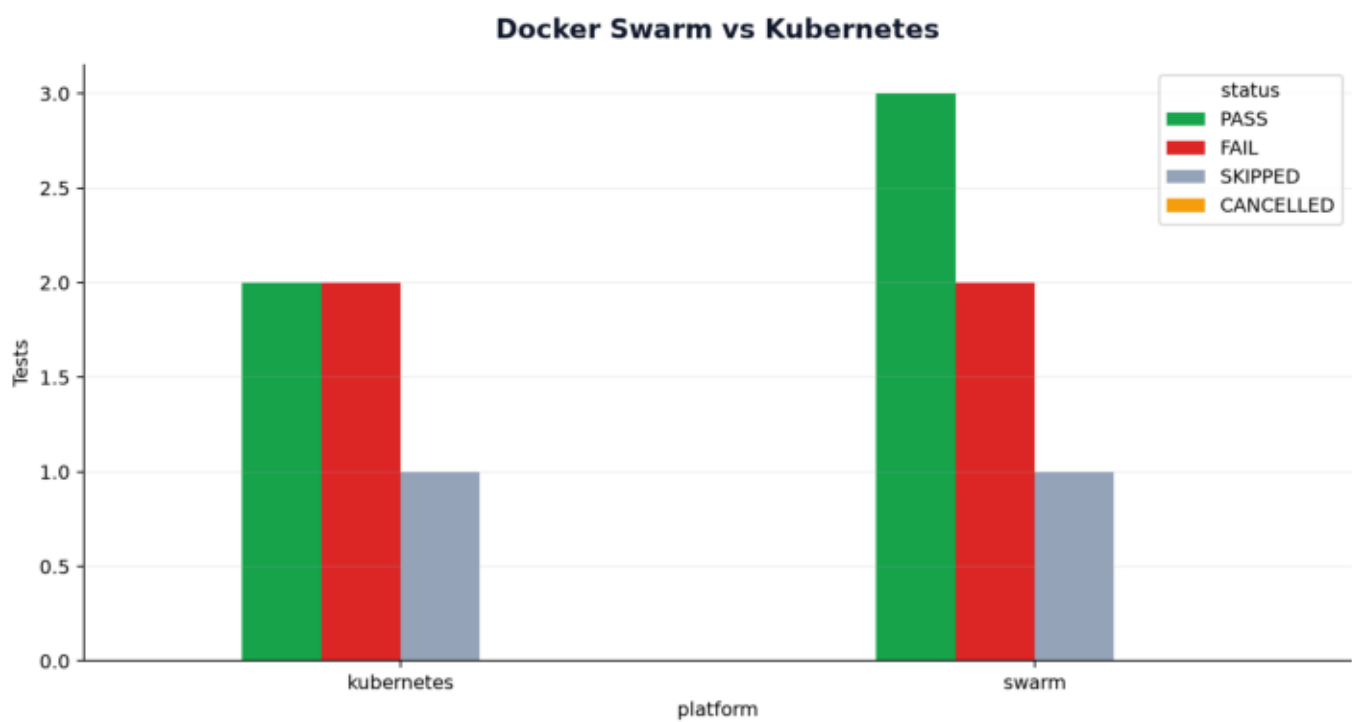
P95 Response Time

Current-session measured data



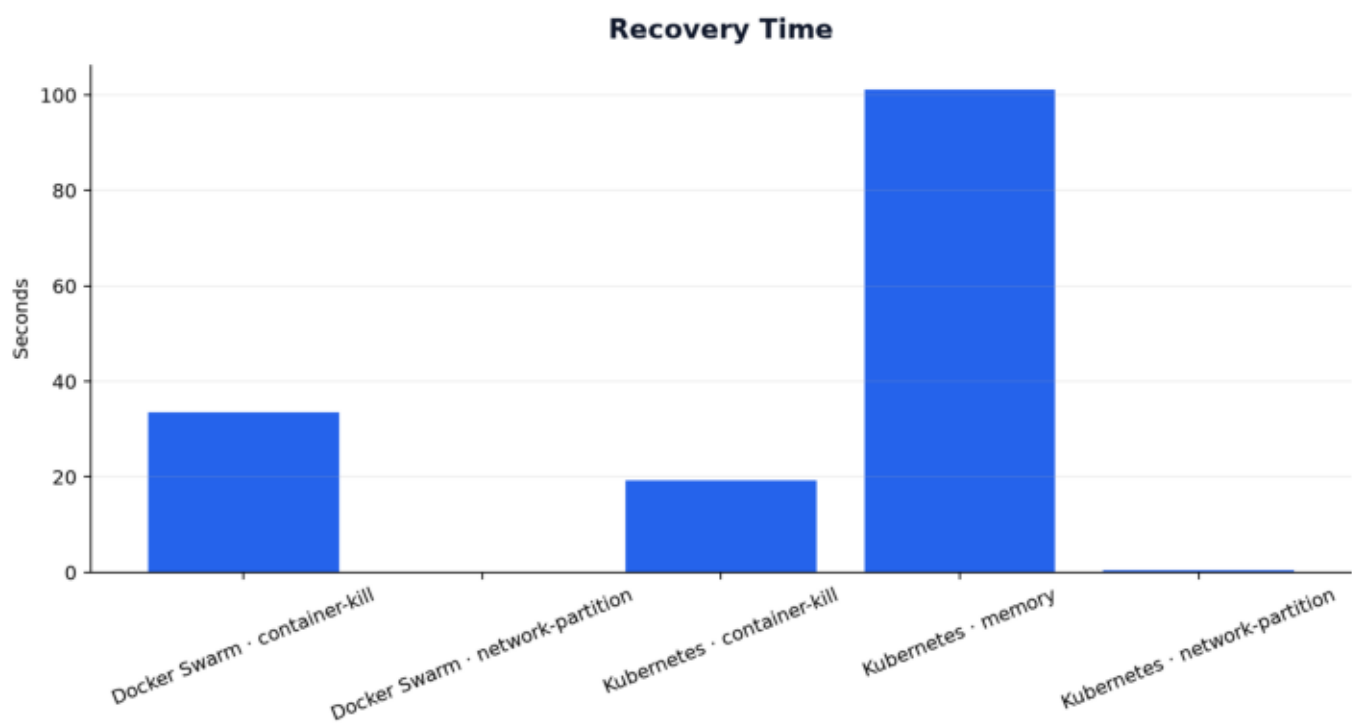
Platform Comparison

Current-session measured data



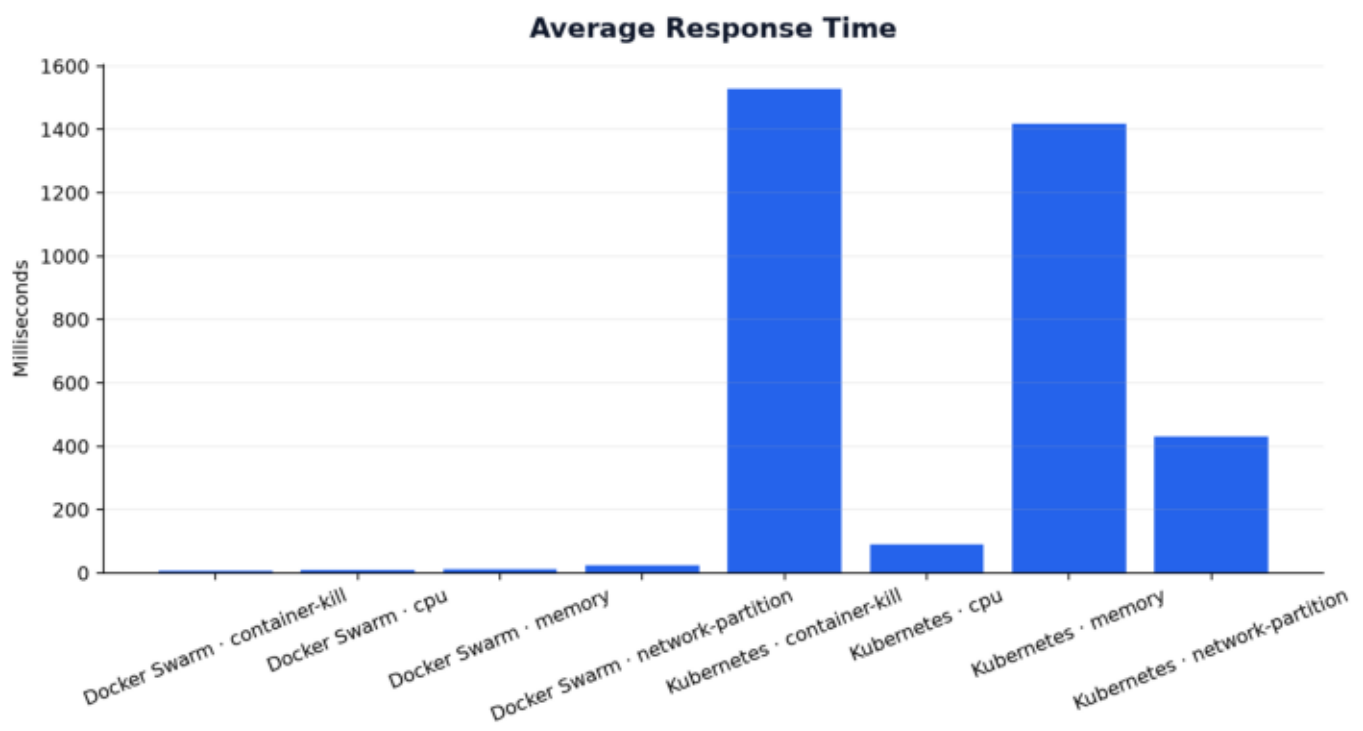
Recovery Time

Current-session measured data



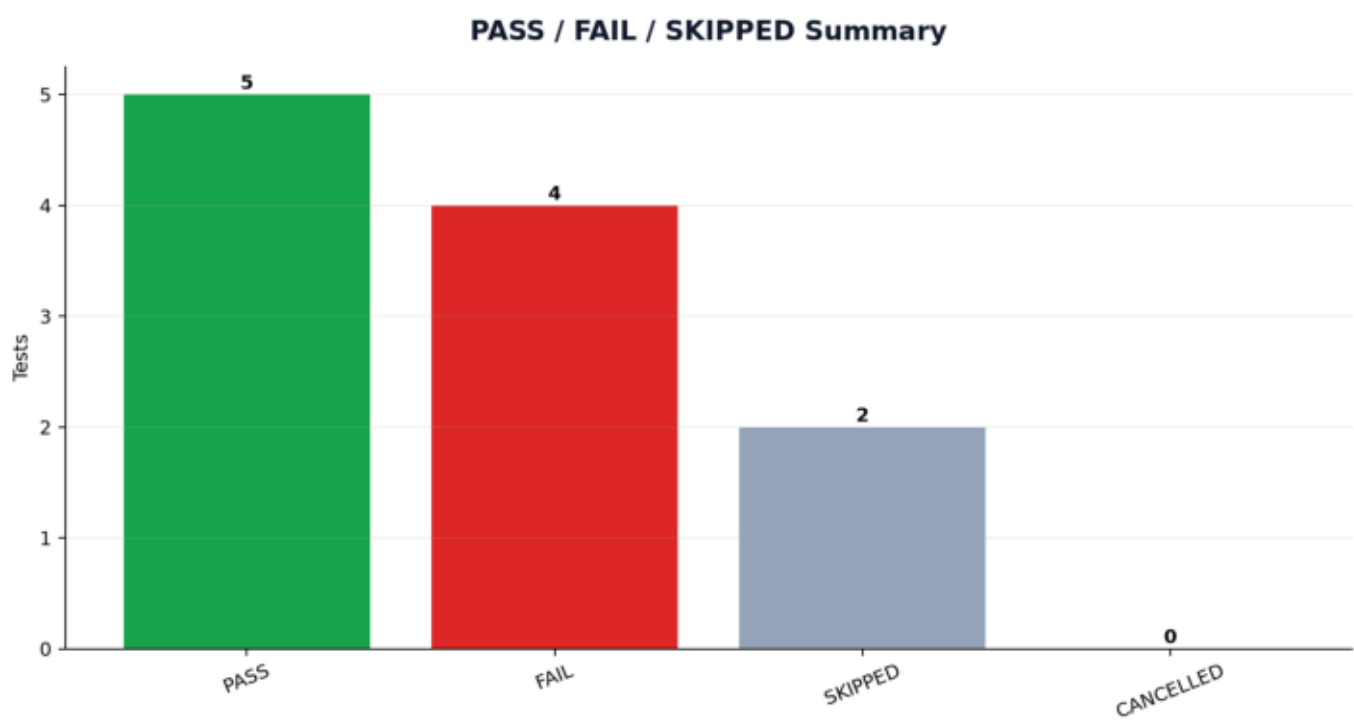
Response Time

Current-session measured data



Status Summary

Current-session measured data



Failures and Errors

Platform	Scenario	Status	Error
swarm	network-partition	FAIL	Critical cleanup/restoration failure: Swarm backend service did not return to Ready state before timeout.
swarm	latency	FAIL	Precondition failed: Swarm backend did not HTTP 200.
kubernetes	memory	FAIL	No healthy backend response was observed during the stress test.
kubernetes	network-partition	FAIL	Critical cleanup/restoration failure: Scenario cancelled by user.

Skipped Tests

Platform	Scenario	Exact reason
swarm	node-failure	Single-node Docker Swarm cannot demonstrate real node rescheduling.
kubernetes	node-failure	Minikube inter-node CNI/DNS does not provide a safe backend-to-PostgreSQL path after rescheduling.

Swarm vs Kubernetes Comparison

Platform	FAIL	PASS	SKIPPED
kubernetes	2	2	1
swarm	2	3	1

Final Conclusion

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During this campaign, 5 of 11 scenarios passed, 4 failed and 2 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.